LQG Control of a Four-Bar Wheeled Bipedal Robot: Toward a Digital Twin, Validated on a CAD-Based Multibody Model.

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***Abstract*—Wheeled bipedal robots (WBRs) with single-actuator four-bar legs can change height with one motor per leg, but they are statically unstable, strongly coupled, and often built with a minimal sensor set. This paper designs and evaluates a discrete-time Linear–Quadratic–Gaussian (LQG) controller for such a robot (WBR-4BL) that uses only five measurements: pitch angle, wheel odometry, heading, and the two leg encoders. A Kalman decomposition of the 15-state linearization shows that the controllable and observable part has exactly nine states, which define the reduced design model. The controller (LQR servo with integral action and a stationary Kalman predictor, 10 ms) is designed on a reduced Lagrangian model. It is validated on an independent multibody digital model generated from the CAD assembly, with compliant wheel–ground contact, actuator dynamics, and sensor noise. We use five scenarios (balance, straight-line, circular, LiDAR-guided tracking, and a ramp with real contact), ten noise realizations each, three baseline controllers, and one-at-a-time robustness sweeps. LQG completed all 50 runs without falling, with a pitch RMSE within 0.12° of that of an ideal state-feedback LQR. The same structure without an estimator fell in every run. The model-based estimator, however, narrows the robustness envelope. The LQG tolerated torque-constant errors down to -10% and body-mass errors of $\pm 10\%$. A gyro-based multi-loop PID tolerated -30% torque error and the full tested mass range (-40% to $+50\%$). Conversely, the LQG tolerated five times the nominal sensor noise, while the PID fell at three times. The digital model is not yet validated against the physical robot, and we state the scope of our claims accordingly.**

***Index Terms*—Wheeled bipedal robot, LQG control, Kalman filter, four-bar mechanism, digital twin, robustness.**

I. Introduction

Wheeled bipedal robots (WBRs) combine efficient rolling with legs that adjust height, cross ramps and reject disturbances [1]–[4]. Like the wheeled inverted pendulum [5], they are open-loop unstable and must modulate wheel torque continuously to keep the center of mass (CoM) above the contact line.

Why a four-bar leg. Table I compares common leg architectures. Fixed legs cannot change height. Serial hip–knee legs need a second motor per leg, usually mounted distally, which raises leg inertia and shifts the wheel fore–aft as the height changes. Parallel five-bar legs [3] keep both motors on the body but still need two actuators per leg. The planar four-bar leg of the WBR-4BL (Fig. 1) uses a *single* hip motor on the body; the legs weigh less than 3% of the robot. With our CAD geometry the wheel moves almost vertically: $|\partial x_w/\partial q| = 0.008$ m/rad versus $|\partial h/\partial q| = 0.171$ m/rad. Height changes therefore barely disturb the pitch balance. The cost is a nonlinear, configuration-dependent map from the hip angle to height and pitch inertia. The wheel motor, mounted on the coupler link, also loads the hip with a reaction torque.

Research gap. Balancing controllers for WBRs with kinematic loops typically use full-state feedback with inertial rate sensing [2] or whole-body/MPC schemes with rich state information [4], [6]. Low-cost platforms, however, often provide only a tilt angle, wheel odometry, a heading estimate and joint encoders. Output-feedback LQG is the natural model-based answer [7], [8], but LQG regulators have no guaranteed stability margins [9]. To our knowledge, the robustness cost of a minimal-sensor LQG on a four-bar WBR has not been quantified, nor compared against simpler alternatives.

Contributions. (i) A discrete LQG for the WBR-4BL whose reduced model is justified by a Kalman decomposition, with explicit controllability and observability analysis. (ii) Validation on a CAD-derived multibody digital model that is independent of the design model, using Monte Carlo statistics over five scenarios. (iii) A quantitative comparison with an LQR without estimator, a gyro-based multi-loop PID and an ideal state-feedback LQR. (iv) A robustness envelope for mass, CoM, friction, actuator and noise variations, which shows the margin lost to the estimator.

TABLE I
Leg architectures for wheeled bipeds

Architecture	Act./leg	Motors	Wheel path (height)
Fixed leg (WIP) [5]	0	–	fixed height
Serial hip–knee	2	hip+knee	arc, fore–aft shift
Parallel five-bar [3]	2	body	any planar (2 DOF)
Four-bar (this work)	**1**	**body**	**near-vertical (1 DOF)**

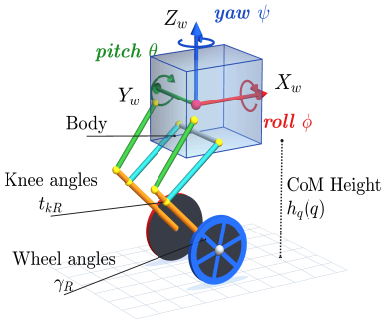


Fig. 1. WBR-4BL: each leg is a planar four-bar linkage driven at the hip by the input (crank) angle t_k, which sets the body height $h(q)$; a hub motor drives each wheel.

II. Robot Platform, Model and Digital Model

A. Platform

The WBR-4BL (Fig. 1) has a total mass of 7.62 kg, of which 7.50 kg is the chassis. Each four-bar leg has a crank of 115 mm, a rocker of 111.3 mm, a coupler of 30 mm and a ground link of 57.7 mm. The usable hip range of $[-0.41, 0.58]$ rad maps to an axle-to-body height of 49–260 mm (197 mm nominal). The wheels have a 50 mm radius and are 122 mm apart. The actuator limits are 20 N m at the hips and 4 N m at the wheels. In the upright equilibrium the hips hold the body weight with -6.37 N m each.

B. Nonlinear model and state reduction

The configuration is described by the planar pose (X, Y, ψ), pitch θ, roll ϕ, the two hip angles (t_{kL}, t_{kR}) and the two wheel angles (γ_L, γ_R). Rolling without slip eliminates the wheel and planar velocities in favor of the forward speed v and yaw rate ω_z. This gives the 15-state model

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}), \quad \boldsymbol{x} = [X, Y, \psi, \theta, \phi, t_{kL}, t_{kR}, \gamma_L, \gamma_R, v, \dot{\theta}, \dot{\phi}, \omega_z, \dot{t}_{kL}, \dot{t}_{kR}]^\top \in \mathbb{R}^{15} \tag{1}$$

with $\boldsymbol{u} = [\tau_L, \tau_R, \tau_{kL}, \tau_{kR}]^\top$ (wheel and hip torques). The design model is a 5-DOF Lagrangian model with $\boldsymbol{Q} = [s, \theta, t_{kL}, t_{kR}, \psi]$. The legs are treated as massless and enter through polynomial fits $h(q)$, $x_w(q)$ of the exact four-bar closure. The roll is fixed by the leg heights, $\phi = (h(t_{kL}) - h(t_{kR}))/b$, and the wheel-motor reaction adds $-\varphi_c'(q)\,\tau_w$ to each hip.

Reduction procedure. The model is linearized at the upright equilibrium by central differences ($\varepsilon = 10^{-6}$). The 15-state linearization $(\boldsymbol{A}_{15}, \boldsymbol{B}_{15}, \boldsymbol{C}_{15})$, with the five measured outputs, is then split by a Kalman canonical decomposition. The controllable subspace has dimension 10 and the observable subspace 9; the minimal realization has 9 states and coincides with

$$\boldsymbol{z} = [\theta, \dot{\theta}, v, \psi, \omega_z, t_{kL}, \dot{t}_{kL}, t_{kR}, \dot{t}_{kR}]^\top. \tag{2}$$

The excluded states are structurally redundant or cannot be reconstructed from the outputs:

- X is controllable but unobservable, since there is no absolute position sensor; the guidance layer closes position through odometry.
- Y is uncontrollable in the linearization, because $\dot{Y} = v \sin\psi$ is nonholonomic.
- $\gamma_{L,R}$ are integrals of v, ω_z, $\dot{\theta}$ and $\dot{t}_k$.
- $(\phi, \dot{\phi})$ are algebraically tied to $t_{kL} - t_{kR}$ ($\partial\phi/\partial(t_{kL} - t_{kR}) = -1.41$).

At the upright equilibrium with the nominal leg angle, the dynamic rows of the reduced model ($\ddot{\theta}$, $\dot{v}$, $\dot{\omega}_z$, $\ddot{t}_{kL}$, $\ddot{t}_{kR}$; the other rows are kinematic) are

$$\boldsymbol{A}_d = \begin{bmatrix} 531.3 & 0 & 0 & 0 & 0 & -13.55 & 0 & -13.55 & 0 \\ -99.58 & 0 & 0 & 0 & 0 & 2.69 & 0 & 2.69 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.73 & 0 & -1.73 & 0 \\ -10.93 & 0 & 0 & 0 & 0 & 44.82 & 0 & -6.66 & 0 \\ -10.93 & 0 & 0 & 0 & 0 & -6.66 & 0 & 44.82 & 0 \end{bmatrix},$$
$$\boldsymbol{B}_d = \begin{bmatrix} -174.7 & -174.7 & -0.57 & -0.57 \\ 35.31 & 35.31 & 0.14 & 0.14 \\ -97.76 & 97.76 & 0.03 & -0.03 \\ 7.86 & 7.15 & 4.84 & 4.00 \\ 7.15 & 7.86 & 4.00 & 4.84 \end{bmatrix} \tag{3}$$

and $\boldsymbol{C}$ selects $(\theta, v, \psi, t_{kL}, t_{kR})$. The entry 531.3 s^{-2} is the gravity term of the inverted pendulum, and the diagonal leg entries 44.82 s^{-2} make the legs unstable under the body weight. The wheel torques act symmetrically on pitch and speed and antisymmetrically on yaw. Through the wheel-motor reaction they also load the hips (first column, last two rows).

Controllability and observability. For the reduced triple $(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C})$, the controllability matrix $[\boldsymbol{B}\ \boldsymbol{AB} \cdots \boldsymbol{A}^8\boldsymbol{B}]$ and the observability matrix $[\boldsymbol{C}; \boldsymbol{CA}; \ldots; \boldsymbol{CA}^8]$ both have full rank 9. The four torques can therefore steer every state of $\boldsymbol{z}$, and the five outputs determine all of them, including the rates $\dot{\theta}$, ω_z and $\dot{t}_k$ that are not measured. Rank is a yes/no test, so we also measure how far each mode is from losing these properties with the Popov–Belevitch–Hautus (PBH) test: $\sigma_{\min}[\lambda_i \boldsymbol{I} - \boldsymbol{A},\ \boldsymbol{B}]$ and $\sigma_{\min}[\lambda_i \boldsymbol{I} - \boldsymbol{A};\ \boldsymbol{C}]$ for every eigenvalue λ_i (SI units, torques in N m). The open-loop eigenvalues are ± 23.06 s^{-1} (pitch, the inverted pendulum), ± 7.18 and ± 6.13 s^{-1} (the antisymmetric and symmetric leg modes, unstable because the hips carry the body weight) and 0 three times (v and the ψ–ω_z double integrator of the heading). The pitch mode is the best controlled (7.95) and the zero modes give 1.0. The smallest values, 0.116 (controllability) and 0.138 (observability), both belong to the antisymmetric leg mode at ± 7.18 s^{-1}. Its input channel is weak: the hip torques enter $\ddot{t}_{kL} - \ddot{t}_{kR}$ with gain $4.84 - 4.00 = 0.84$ (Eq. (3)), against 8.84 for the sum, because each hip torque also loads the opposite leg through the body. This mode also sets the roll, and it is the one that grows before the falls of the robustness study (Sec. V-C).

C. Multibody digital model (validation plant)

The validation plant is a Simscape Multibody model generated by script from the CAD assembly. It uses the STEP geometry and the exported mass properties of nine bodies, ten joints and the closed four-bar loops. It is *not* the design model:

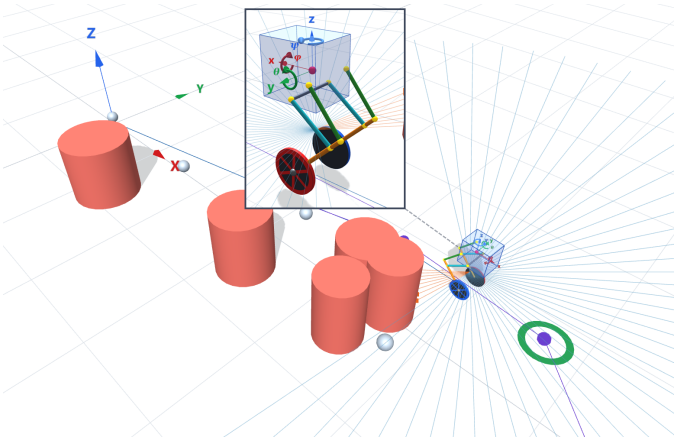


Fig. 2. Browser simulator of the digital model in the LiDAR scenario with random obstacles ($t = 10.9$ s): A* path (violet), waypoints (white), LiDAR beams (blue; orange on hits) and goal (green ring). Inset: the robot, drawn from the same CAD geometry as Fig. 1, with its body frame and the roll ϕ, pitch θ and yaw ψ axes.

link masses, contact compliance and actuator dynamics are all present. The wheel–ground, wheel–ramp and chassis–obstacle contacts use a penalty model ($k = 2{\cdot}10^5$ N/m, $b = 10^3$ N s/m) with smooth stick–slip friction ($\mu_s/\mu_k = 0.9/0.7$). Each motor has its rotor inertia and a 2-ms current loop. The controller runs as a 10-ms discrete block. A guidance layer provides pure-pursuit speed and heading references and a 2-D LiDAR with an A* planner [10]. All sensor noise is seeded, so every run is reproducible.

Scope of the term "digital twin". Following the classification of [11] (see also [12]), our environment is a *digital model*: there is no automatic data exchange with the physical robot yet, and it has not been validated against hardware measurements. It therefore serves as a virtual prototype. The evidence we provide is (a) model-to-model consistency, since the same synthesis code reproduces the gains of an independent browser simulator of the robot to a relative error of $1.2{\cdot}10^{-5}$ (Fig. 2), and (b) controller behavior under deliberate mismatch between the design model and the plant. We do not claim correspondence with the physical platform; that is left to future work.

III. LQG Controller Design

Fig. 3 shows the resulting architecture. The continuous pair $(\boldsymbol{A}, \boldsymbol{B})$ is discretized by zero-order hold [13] with $h = 10$ ms, giving $(\boldsymbol{\Phi}, \boldsymbol{\Gamma})$. Integral states $\boldsymbol{\xi}_{k+1} = \boldsymbol{\xi}_k + h(\boldsymbol{r}_k - \boldsymbol{C}_s\hat{\boldsymbol{z}}_k)$ act on the servo outputs $\boldsymbol{C}_s\boldsymbol{z} = [v, \psi, t_{kL}, t_{kR}]^\top$. The LQR gain $\boldsymbol{K}_a = [\boldsymbol{L}_z \ \boldsymbol{L}_i]$ minimizes $\sum(\boldsymbol{z}_a^\top \boldsymbol{Q}\boldsymbol{z}_a + \boldsymbol{u}^\top \boldsymbol{R}\boldsymbol{u})$ for the augmented plant. A stationary one-step Kalman predictor

$$\hat{\boldsymbol{z}}_{k+1} = \boldsymbol{\Phi}\hat{\boldsymbol{z}}_k + \boldsymbol{\Gamma}\boldsymbol{u}_k + \boldsymbol{K}_f(\boldsymbol{y}_k - \boldsymbol{C}\hat{\boldsymbol{z}}_k) \tag{4}$$

uses the five noisy measurements $\boldsymbol{y} = [\theta, v, \psi, t_{kL}, t_{kR}]^\top + \boldsymbol{v}$. By the separation principle, the control law is

$$\boldsymbol{u}_k = \boldsymbol{u}_{eq} + \boldsymbol{u}_{ff}(\alpha) - \boldsymbol{L}_z(\hat{\boldsymbol{z}}_k - \boldsymbol{z}_{ff}(\alpha)) - \boldsymbol{L}_i\boldsymbol{\xi}_k, \tag{5}$$

with torque saturation and integrator anti-windup. Here $\boldsymbol{u}_{eq}$ is the hip gravity compensation, and $(\boldsymbol{z}_{ff}, \boldsymbol{u}_{ff})$ is the static equilibrium on a slope α, tabulated from the design model. On slopes the same synthesis is repeated every 2° between $\pm 16^\circ$ and the resulting gains are interpolated. This is needed because the flat-ground LQG becomes linearly unstable beyond $\pm 8^\circ$ through a gravity-induced yaw mode. Table II lists the design values. The closed loop has $\rho(\boldsymbol{\Phi}_a - \boldsymbol{\Gamma}_a\boldsymbol{K}_a) = 0.9976$ and the estimator $\rho(\boldsymbol{\Phi} - \boldsymbol{K}_f\boldsymbol{C}) = 0.850$; the scheduled design stays below 0.998 on all slopes.

TABLE II
LQG design parameters

h / (n, m, p, n_i)	10 ms / (9, 4, 5, 4)
$\boldsymbol{Q}_z$	diag(500,5,50,300,30,20,4,20,4)
$\boldsymbol{Q}_i$ / $\boldsymbol{R}$	diag(40,150,15,15) / $0.0666\,\boldsymbol{I}_4$
$\boldsymbol{Q}_n \cdot 10^4$	diag(10^{-3},2,4,10^{-3},2,10^{-3},5,10^{-3},5)
σ of θ, v, ψ, t_k	0.11°, 2.5 mm/s, 0.17°, 0.11°
Fall criterion	$\lvert\theta\rvert > 34^\circ$ or $\lvert\phi\rvert > 42^\circ$

IV. Validation Methodology

Scenarios. The scenarios are:

- S1, *balance*: 8 s in place from a 2° initial tilt.
- S2, *forward*: a 3-m point-to-point run (14 s).
- S3, *circle*: closed-loop tracking of a circle of radius 1.2 m (25 s).
- S4, *obstacle*: LiDAR/A* navigation past two cylinders to $(3.2, 0)$ m (18 s).
- S5, *ramp*: ascent, plateau and descent of a 15-cm ramp (maximum slope 12°) with real contact, ending at 5.85 m (22 s).

Statistics. Each scenario is repeated with ten independent noise seeds. We report the mean $\pm$ standard deviation of the pitch, roll and speed-error RMSE, the peak pitch, the control energy $\int \|\boldsymbol{u}\|^2 dt$ (which includes the hip gravity torques) and the final goal error.

Baselines. All baselines use the same gains, saturations, guidance and slope feedforward:

- *LQR + finite differences*: the same five sensors, with rates from filtered differences ($T_f = 5$ ms) instead of the Kalman predictor.
- *Multi-loop PID + gyro*: four decoupled SISO loops (pitch–speed, heading, leg height, roll/leg difference) obtained by projecting the LQR gain onto modal channels and discarding all cross-channel terms [14]. It also reads a rate gyro (θ, ω_z; $\sigma = 0.29^\circ$/s).
- *Ideal LQR*: the true, noise-free state, as a performance bound.

Robustness. The design is kept nominal while the plant is perturbed, one parameter at a time, in S2 with three seeds: torque constant k_τ, body mass k_m (scaling mass and inertia), CoM offsets, friction, current-loop lag and noise level. The closed-loop spectral radius with the perturbed linearized plant gives the corresponding linear stability limits.

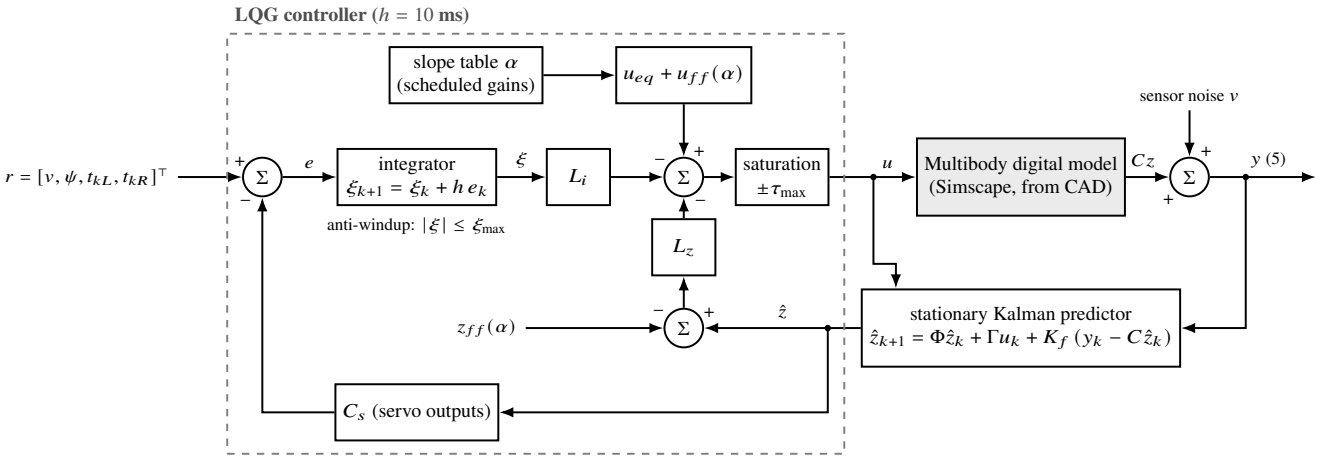


Fig. 3. Control architecture. The servo integrates the error between the references $\boldsymbol{r}$ and the estimated servo outputs $\boldsymbol{C}_s\hat{\boldsymbol{z}}$; the Kalman predictor reconstructs $\hat{\boldsymbol{z}}$ from the saturated torques $\boldsymbol{u}$ and the five noisy outputs $\boldsymbol{y}$; the slope table supplies $\boldsymbol{u}_{eq} + \boldsymbol{u}_{ff}(\alpha)$, $\boldsymbol{z}_{ff}(\alpha)$ and the scheduled gains. The plant is the multibody digital model, not the design model.

V. Results and Discussion

A. Nominal performance (Monte Carlo)

Table III and Fig. 4 summarize the LQG results. The controller completed *all* 5×10 runs without falling.

S1 (balance). The pitch RMSE is $0.29 \pm 0.02^\circ$ including the initial 2° transient (Fig. 4, row 1). The Kalman estimate follows the true pitch within the sensor noise, and the hip torques settle at the gravity-compensation level. The roll RMSE is essentially a static lean (mean 1.18°). It appears with every controller, because the servo enforces equal leg angles rather than a level body.

S2 (forward). The robot reaches the 3-m goal with a final error of 0.20 m. The speed RMSE (0.335 m/s) is dominated by the commanded acceleration and braking transients, because the speed reference enters only through the integral state. This produces a small overshoot and back-up at the goal (row 2).

S3 (circle). Sustained turning keeps the pitch RMSE at 0.45°, with a roll RMSE of 2.69°. The robot enters the circle with a larger radius and then follows it in steady state (row 3). Heading and balance are thus effectively decoupled under persistent lateral excitation.

S4 (obstacle). The LiDAR-guided reference keeps a mean minimum clearance of 0.118 m (worst 0.118 m) with 0 collisions, at a pitch RMSE of 0.82° (row 4).

S5 (ramp). This is the most demanding case. The peak pitch of 7.2° occurs at the grade transitions, where the slope feed-forward and the scheduled gains switch between equilibria. Roll stays bounded while the legs absorb the uneven contact, and the robot ends 0.19 m from the goal (row 5).

B. Comparison with alternative controllers

The right half of Table III compares the controllers. Without an estimator, the same LQR fed by finite differences of the five measurements falls in 50/50 runs, always within 0.28 s. The 23-s^{-1} unstable pitch pole cannot be stabilized with the delay and noise of differentiated tilt data at 10 ms, so the Kalman predictor is *necessary* for this sensor set. With only the five measurements, the LQG stays within 0.12° of the pitch RMSE of the ideal LQR in every scenario, at a similar control energy. The gyro-based multi-loop PID is practically indistinguishable from the ideal LQR (within 0.01°). A rate gyro removes the need for model-based rate reconstruction, and the cross-channel terms of $\boldsymbol{L}_z$ add little at this operating point. In nominal conditions, then, the LQG pays only a small accuracy penalty for working without rate sensors.

C. Robustness envelope

Fig. 5 shows the robustness sweeps. The linear analysis predicts closed-loop stability for torque factors $k_\tau \in [0.89, 1.46]$ and body-mass factors up to $k_m = 1.10$ with the LQG. The ideal LQR and the multi-loop PID admit $k_\tau \in [0.51, 1.86]$ and k_m up to 1.95. The multibody simulations confirm both LQG limits: it survives $k_\tau = 0.90$ and $k_m = 1.10$ but falls at $k_\tau = 0.85$ and $k_m = 1.15$. The model is optimistic for a lighter body. At $k_m = 0.8$ the LQG is linearly stable but diverges slowly (a fall at 7.7 s) through a growing leg–roll oscillation with saturated hip torques. The LQG also falls with fore–aft CoM offsets of ± 20 mm and a 20-ms current lag. It is insensitive to vertical CoM offsets (-20 to +40 mm) and to friction down to $\mu_s = 0.15$. The PID with gyro survives every mass, CoM, friction and lag case and falls only at $k_\tau = 0.6$. The trend reverses for sensor noise: the LQG survives five times the nominal noise, whereas the PID falls at three times, because its hip rates come from differentiated encoder signals.

The margin loss is the classical LQG effect [9]. The predictor integrates the nominal $\boldsymbol{\Gamma u}$ through the fast unstable pitch mode, so an error in torque gain or inertia becomes a biased rate estimate that the regulator then amplifies. Our checks agree: the ideal LQR with the same $\boldsymbol{L}_z$ keeps the wide margins, and in a preliminary single-seed test, adding the gyro to the Kalman filter alone did not restore them. A fixed-gain LQG with this minimal sensor set is therefore adequate only inside the identified envelope. On hardware, the

TABLE III

CLOSED-LOOP PERFORMANCE OVER 10 NOISE SEEDS PER SCENARIO: LQG (MEAN ± S.D.) AND ALTERNATIVE CONTROLLERS (MEANS OVER NON-FALLING RUNS; IDEAL LQR: ONE NOISE-FREE RUN)

	LQG, 5 outputs (proposed)							LQR + finite diff.		Multi-loop PID + gyro			Ideal LQR	
Scen.	RMSE θ (°)	max $\lvert\theta\rvert$ (°)	RMSE ϕ (°)	RMSE e_v (m/s)	Energy (N^2m^2s)	Goal err. (m)	Falls	Falls	t_{fall} (s)	RMSE θ (°)	Energy (N^2m^2s)	Falls	RMSE θ (°)	Energy (N^2m^2s)
Balance	0.29 ± 0.02	2.0	1.27 ± 0.00	0.021 ± 0.001	656	0.010 ± 0.004	0/10	10/10	0.24	0.17	674	0/10	0.17	652
Forward	1.44 ± 0.01	3.4	1.11 ± 0.00	0.335 ± 0.001	1039	0.196 ± 0.006	0/10	10/10	0.25	1.41	1083	0/10	1.40	1048
Circle	0.45 ± 0.01	2.0	2.69 ± 0.01	0.093 ± 0.001	2184	—	0/10	10/10	0.23	0.37	2161	0/10	0.38	2098
Obstacle	0.82 ± 0.01	2.0	1.26 ± 0.01	0.197 ± 0.002	1477	0.144 ± 0.005	0/10	10/10	0.23	0.79	1516	0/10	0.78	1469
Ramp	1.69 ± 0.01	7.2	1.50 ± 0.08	0.296 ± 0.001	1374	0.191 ± 0.007	0/10	10/10	0.25	1.62	1415	0/10	1.61	1369

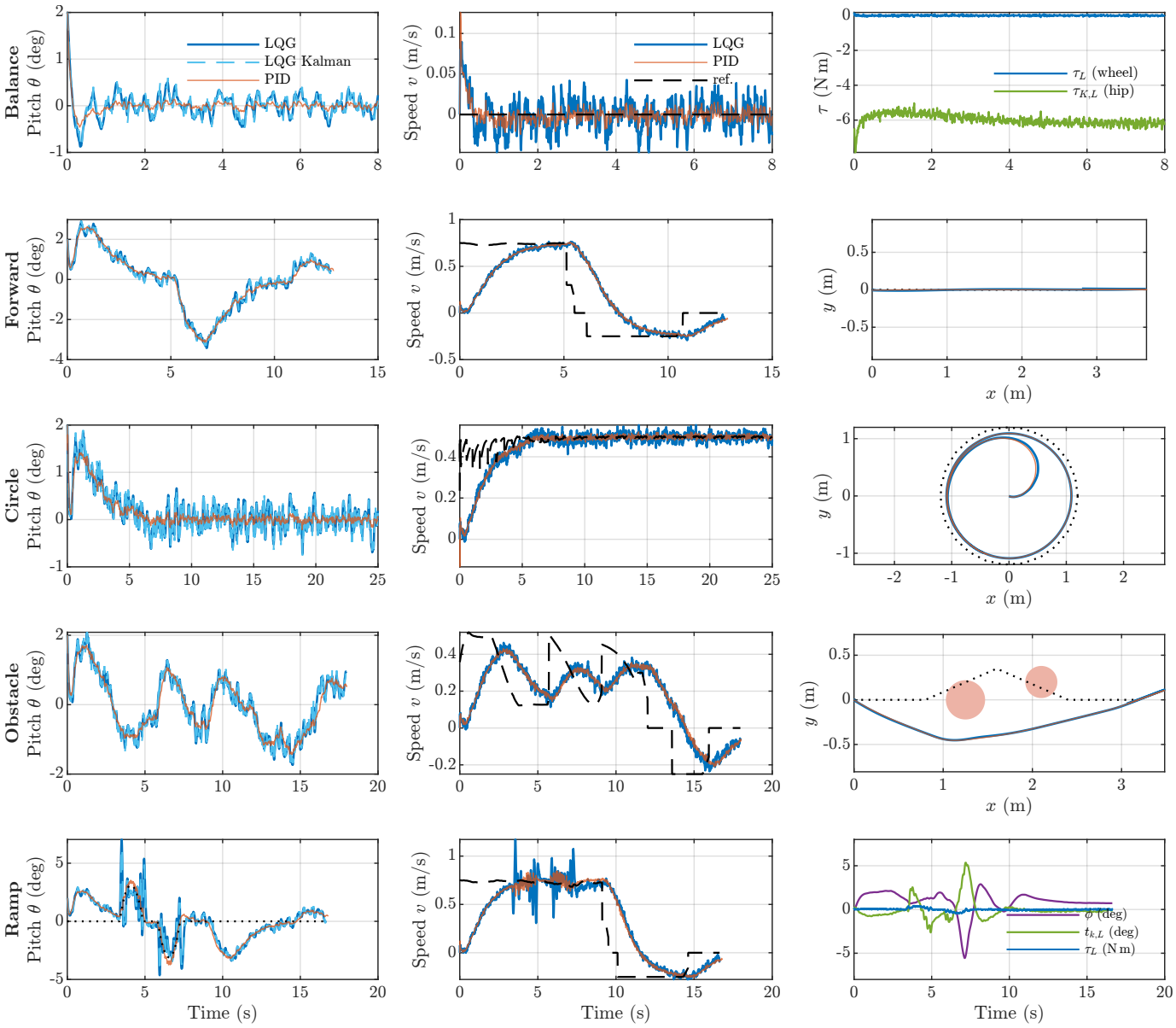


Fig. 4. Representative runs (seed 7) for S1–S5 (rows). Left: pitch, true (blue) and Kalman estimate (cyan, dashed), with the multi-loop PID in orange and the slope feedforward θ_{ff} (dotted) in S5. Center: forward speed and reference. Right: S1, wheel and hip torques; S2–S4, path, waypoints and obstacles; S5, roll, leg angle and wheel torque.

rate gyro present in any IMU should be used directly in the loop, or loop-transfer recovery [15] should be applied, and the torque constant and mass should be identified to within about 10%. Finally, the deliberate failure scenarios of the browser study (an impossibly steep ramp, ice, a strong impact and weakened actuators) also fall in the multibody model. They

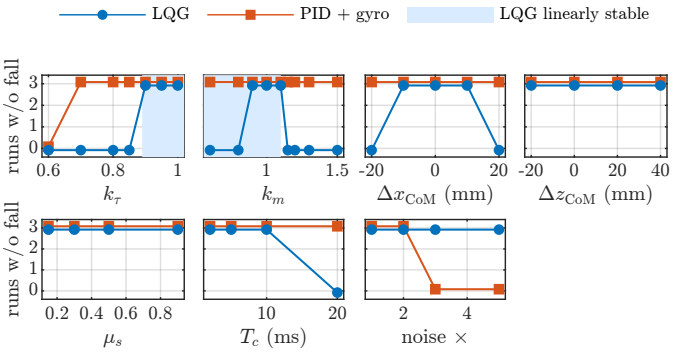


Fig. 5. Robustness envelope in S2: number of runs (out of 3 seeds) without fall versus each plant perturbation, for LQG (blue) and multi-loop PID with gyro (orange). The shaded band marks the linear stability region of the LQG.

confirm that these limits are physical (friction cone, torque, region of attraction) rather than algorithmic.

VI. Conclusion

We presented a discrete LQG controller for a single-actuator four-bar wheeled biped that uses only five measurements. Its nine-state design model follows from a Kalman decomposition of the 15-state linearization. Validated over 50 Monte Carlo runs on an independent CAD-derived multibody model, it never fell in five scenarios and stayed within 0.12° of the pitch RMSE of an ideal state-feedback LQR. The same structure without an estimator failed in every run, so the Kalman predictor is essential for this sensor set. The robustness study gives the price of that estimator: about 10% tolerance in torque gain and body mass (linear limits of 11% and 10%), against 30% and more than 40% for gyro-based feedback. In exchange, the LQG filters sensor noise much better. These results are model-based; future work will validate the digital model against the physical robot, add the IMU rate gyro to the estimator, and apply loop-transfer recovery or adaptive identification of the actuator gain.

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